# A MODERN BAG OF TRICKS FOR MULTI-CAMERA RE-IDENTIFICATION

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## ABSTRACT

Person and object re-identification (ReID) requires embeddings that stay discriminative across cameras, viewpoints, and partial occlusions. We present a training and inference pipeline for multi-camera ReID built on DINOv2, a self-supervised Vision Transformer. The model concatenates the global class token with averaged patch tokens, so the embedding keeps both semantic context and local appearance, and is trained with a hybrid loss and a BNNeck. For tracklet-level matching, we compare density-based clustering with Generalized Mean (GeM) pooling for aggregating frame embeddings, and refine the ranking with $k$-reciprocal re-ranking. Since AICity/MTMC Tracking 2025 has no official ReID split, we propose a one-query-per-identity tracklet evaluation protocol for AI City/MTMC Tracking 2025, adapting Market-1501's junk-identity convention to a dataset that otherwise has no standardized ReID split. The pipeline reaches **94.2% mAP** on Market-1501, competitive with other state-of-the-art methods, and **91.6% mAP** on the proposed AI City protocol (longest tracklet as query), **+9.3 mAP** over the fully trained baseline. The pipeline is fully reproducible and shows that a self-supervised backbone, combined with several training tricks, can improve multi-camera ReID. The code is available on GitHub.

## 1 INTRODUCTION

Person and object re-identification (ReID) is a core retrieval problem in computer vision. The task is to match the same identity across a network of distributed, non-overlapping cameras. That requires embeddings that remain discriminative under viewpoint change, illumination variation, background clutter, and partial occlusion. Weak or domain-sensitive embeddings produce incorrect cross-camera matches (Hashempoor, 2025). In this work, we fine-tune feature-extraction backbones, compare tracklet clustering and pooling strategies, and apply Random Erasing for reliable cross-camera identity retrieval. The focus is on the re-identification of people and objects in the AI City / MTMC Tracking 2025 dataset.

Historically, Convolutional Neural Networks (CNNs) such as ResNet (He et al., 2016) and OSNet (Zhou et al., 2019) dominated ReID feature extraction. While effective, their limited receptive fields restrict the modeling of long-range spatial relationships. Vision Transformers (ViTs) (Dosovitskiy et al., 2020) overcome this limitation by capturing global context, and self-supervised models such as DINOv2 (Oquab et al., 2023) learn richer visual representations. However, directly applying DINOv2 to ReID remains challenging due to domain shifts, viewpoint and illumination variations, background clutter, and occlusions.

To address these challenges, we adapt the self-supervised Vision Transformer DINOv2 for fine-grained ReID by combining class and patch-average token representations with a hybrid multi-task loss. We further improve temporal tracklet aggregation by replacing DBSCAN-based pooling with Generalized Mean (GeM) pooling and applying $k$-reciprocal re-ranking. This combination yields substantial gains over a fully fine-tuned CNN baseline under the same training recipe (+9.3 mAP on AI City/MTMC Tracking 2025) while remaining competitive with person-pretrained state-of-the-art methods on Market-1501.

## 2 BACKGROUND

Let $\mathcal{X}$ denote the set of the cropped object images from input video frames, $\mathcal{Y} = \{1, \ldots, N_y\}$ the ground-truth identity labels, and $\mathcal{C} = \{c_1, \ldots, c_M\}$ a network of non-overlapping camera views. Given a query instance $q \in \mathcal{X}$ with label $y_q$ captured by camera $c_q$, ReID retrieves matching identity instances from gallery $\mathcal{G} = \{(g_j, y_{g_j}, c_{g_j})\}_{j=1}^{N_g}$ using a distance metric $d(\phi(q), \phi(g_j))$ over feature extractor $\phi : \mathcal{X} \to \mathbb{R}^D$. In image-level ReID, queries and gallery samples are single crops ($q, g_j \in \mathcal{X}$). In tracklet-level ReID, queries $T_Q = \{q_k\}_{k=1}^{K_Q}$ and gallery candidates $T_{G,j} = \{g_{j,m}\}_{m=1}^{K_{G,j}}$ are tracklets, where a tracklet is defined as $T = \{x_k\}_{k=1}^{K}$ , a temporal sequence of $K$ cropped image frames corresponding to a single target identity observed continuously within one camera.

The frame-level embeddings $\phi(x_k) \in \mathbb{R}^D$ within a tracklet are aggregated into a single tracklet descriptor via temporal pooling $\Psi : \mathbb{R}^{K \times D} \to \mathbb{R}^D$:

$$F_T = \Psi\left(\{\phi(x_k)\}_{k=1}^{K}\right) \in \mathbb{R}^D. \tag{1}$$

Cross-camera matching is then evaluated directly on aggregated tracklet descriptors of query ($F_Q$) and gallery ($F_G$) using cosine distance. The resulting distance matrix may optionally be refined using $k$-reciprocal re-ranking during post-processing.

## 3 RELATED WORK

Early ReID approaches relied on handcrafted descriptors such as LOMO (Liao et al., 2015) and ELF (Gheissari et al., 2006), combined with metric learning methods including XQDA (Liao et al., 2015) and KISSME (Koestinger et al., 2012). The emergence of deep CNN-based models, such as ResNet-50 (He et al., 2016) and OSNet (Zhou et al., 2019), enabled hierarchical feature learning and significantly improved ReID performance. In particular, OSNet introduced omni-scale feature learning to capture features at multiple spatial scales. Transformer-based architectures (Vaswani et al., 2017) have further advanced ReID by modeling global relationships through self-attention, resulting in more robust feature representations across camera views (Bhuiyan et al., 2025). Vision Transformer (ViT) models, including TransReID (He et al., 2021) and DINOv2 (Oquab et al., 2023), leverage global context to improve feature discrimination. However, directly applying these large foundation models to the fine-grained identity discrimination remains challenging due to domain mismatch, leading to the development of ReID-specific architectural adaptations (Luo et al., 2019) and parameter-efficient fine-tuning techniques such as LoRA (Hu et al., 2022).

Metric learning has become a fundamental component of modern person ReID by learning embedding spaces in which samples of the same identity are grouped together while different identities are well separated. Most competitive ReID frameworks, including Bag of Tricks (Luo et al., 2019) and TransReID (He et al., 2021), adopt a hybrid optimization strategy that combines Cross-Entropy (Label-Smoothed CE in (Luo et al., 2019)) and Batch-Hard Triplet loss as the standard training objective. More recent approaches built upon Vision Transformer foundation models further incorporate objectives such as Supervised Contrastive Loss (Khosla et al., 2020) and ArcFace (Deng et al., 2019) to improve embedding compactness and inter-class discrimination. These complementary losses jointly optimize classification performance and metric representation learning. Prior AI City work on vehicle track retrieval similarly combines multiple objectives: intra/inter-modal, attribute, and cycle-consistency losses with tuned weights for text-to-image rather than purely visual matching (Sebastian et al., 2021).

Tracklet aggregation aims to combine multiple frame-level embeddings from the same tracklet into a single discriminative tracklet representation. Early aggregation strategies primarily relied on simple mean and max pooling (Gholamalinezhad & Khosravi, 2020), which provide computational efficiency but treat all frames equally or rely solely on the strongest activations, making them less robust to noisy or occluded data.

To improve the quality of tracklet representations, clustering-based aggregation methods have been introduced. The baseline GLANCE framework (Hashempoor, 2025) employs DBSCAN to filter noisy frame embeddings before cross-camera association. More generally, DBSCAN (Ester et al., 1996) and its hierarchical extension HDBSCAN (Campello et al., 2013) identify dense regions in the embedding space while rejecting outliers caused by motion blur, occlusion, or inaccurate detections. Alternatively, Generalized Mean (GeM) pooling (Radenović et al., 2018) introduces a learnable power parameter that interpolates between average and max pooling, enabling more discriminative aggregation of frame-level features. Beyond feature aggregation, retrieval performance can be further improved through manifold refinement via $k$-reciprocal re-ranking (Zhong et al., 2017).

# 4 METHOD

We evaluate our framework on two complementary datasets to validate large-scale generalization and enable comparison with prior work.

**AI City / MTMC Tracking 2025:** A multi-camera synthetic benchmark containing multi-camera video sequences across multiple scenes (Tang et al., 2025), (Wang et al., 2024a), (Wang et al., 2024b). Serves as the primary dataset for this implementation.

**Market-1501:** A benchmark dataset widely adopted for person re-identification, captured in real-world scenarios (Zheng et al., 2015). Used to validate the pipeline and methods on a well-established ReID dataset.

## 4.1 MODEL ARCHITECTURE

The proposed ReID framework uses a modular architecture comprising a backbone feature extractor, embedding head, and classification head. This allows different backbones to be evaluated within a common training and inference pipeline. We evaluate the following models:

- DINOv2 (`dinov2_vitb14`), Vision Transformer, pretrained on the LVD-142M dataset
- OSNet (`osnet_x1_0`), trained on Market-1501 dataset
- OSNet-AIN (`osnet_ain_x1_0`), trained on MSMT17, DukeMTMC-reID and CUHK03 datasets.

Given an input image $X$, the DINOv2 backbone produces normalized class token representation $x_{\text{cls}} \in \mathbb{R}^d$ and patch token representations $X_{\text{patch}} = \{x_{\text{patch},i}\}_{i=1}^{N_p} \subset \mathbb{R}^d$, where $d = 768$ for ViT-B/14 and $N_p$ is the number of non-overlapping $14 \times 14$ patches. The class token captures global semantic information, whereas the averaged patch embeddings preserve local spatial appearance information and concatenating both of these results in 1536-D feature vector $f_{\text{cat}}$.

$$f_{\text{cat}} = \left[ x_{\text{cls}} \,\|\, \frac{1}{N_p} \sum_{i=1}^{N_p} x_{\text{patch},i} \right] \in \mathbb{R}^{2d}. \tag{2}$$

The concatenated feature vector $f_{\text{cat}}$ is subsequently passed into a Batch Normalization Neck (`BNNeck`) (Luo et al., 2019). The `BNNeck` applies 1D Batch Normalization that acts on each feature dimension separately and standardizes the distribution of that feature and helps in stabilizing training.

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \tag{3}$$

Batch Normalization is followed by L2-normalization on image's entire embedding and standardizes the vector length to 1, so that similarity computation depends only on direction and not on magnitude of the vector. During training, a linear classifier $W \in \mathbb{R}^{N_{\text{classes}} \times D}$ projects the final embedding vector $f_{\text{emb}}$ (1536-D) to identity classification logits $z = W f_{\text{emb}}$.

To examine the effect of parameter-efficient adaptation versus full model adaptation of DINOv2, our model architecture supports both full fine-tuning (unfreezing all ViT backbone parameters) and Low-Rank Adaptation (`LoRA`) (Hu et al., 2022). In the LoRA configuration, all the backbone weights are frozen, and rank-$r$ parameter update matrices with rank $r \in \{8, 16, 32\}$.

$$W + \Delta W = W + \frac{\alpha}{r} BA \tag{4}$$

These matrices are injected into the query and value projections ($Q$ and $V$) within the backbone's multi-head self-attention modules.

### 4.2 Training Strategy

The training pipeline is engineered for object re-identification, incorporating data preprocessing, data augmentations, identity-balanced batch sampling, and mixed-precision training.

**Data Preprocessing and Splitting:** For AI City / MTMC Tracking 2025 dataset, raw video sequences are preprocessed into frames by using bounding box information and subsequently grouped into tracklets. Extracting all frames from videos of AI City results in millions of images, therefore we subsample annotations at 1 FPS (annotation stride of 30 frames at 30 fps). Tracklets are split whenever temporal gaps exceed 30 frames, and tracklets containing fewer than 2 frames are discarded. If a tracklet exceeds 30 frames, 30 evenly spaced frames are uniformly sampled. For cross-camera evaluation, identities appearing across at least two distinct cameras are retained and others are discarded. Preprocessed tracklets are filtered and we use a one-query-per-identity protocol in the spirit of DukeMTMC-VideoReID (Wu et al., 2018), since the dataset did not have a pre-defined query and gallery splits. The longest tracklet of an identity is designated as the Query, while all remaining tracklets of that identity from different cameras form the Gallery. The resulting splits are summarized in Table 1. The training split contains 862 identities observed across 347 cameras, while evaluation is restricted to 131 cross-camera identities in 67 cameras. Since the longest tracklet of each identity is designated as the query, the query set contains exactly one tracklet per identity. One identity was excluded because its only tracklet contained just two frames, yielding 993 usable identities in total ($862 + 131$). Notably, the query set is much smaller than the gallery (131 vs. 7,503 tracklets), so each query retrieves against a large and diverse candidate gallery. Note that the test set from the dataset is excluded as it does not provide any labels for the data.

Table 1: AI City / MTMC Tracking 2025 split after preprocessing. Note that the query and gallery have the same number of identities as it is taken from the val set of AI City / MTMC Tracking 2025.

| **Split** | **Images** | **Tracklets** | **Cameras** | **Identities** |
|---|---|---|---|---|
| Train | 1,214,776 | 79,622 | 347 | 862 |
| Query | 3,925 | 131 | 67 | 131 |
| Gallery | 106,374 | 7,503 | 67 | 131 |
| Total | 1,325,075 | 87,256 | 414 | 993 |

For Market-1501 dataset, no preprocessing was done so that the results from this pipeline could be comparable with other state-of-the-art results. The test set of Market-1501 dataset comes with a pre-defined query and gallery splits for benchmarking.

**Data Augmentation & Input Pipeline:** Input images are resized to a high resolution of $384 \times 192$ (height $\times$ width) in the final setting. The training pipeline applies 10-pixel padding followed by random horizontal flipping ($p = 0.5$), color jittering (brightness, contrast, and saturation factors of 0.2), and ImageNet mean/standard normalization. Additionally, Random Erasing randomly masks rectangular regions of the input image to simulate partial occlusions and forces the network to learn robust object representations from multiple regions instead of relying only on a single discriminative feature (Luo et al., 2019). Evaluation pipeline uses only deterministic resizing and normalization.

**Identity-Balanced Sampling and Optimization:** Mini-batches for training are constructed using a PK sampling strategy (`PKSampler`), where each batch containing $P = 16$ unique identities and $K = 4$ images per identity, creating a mini-batch size of $B = P \times K = 64$. This guarantees the presence of positive and negative examples required by metric learning losses while maintaining balanced identity representation throughout training (Luo et al., 2019). When an identity contains fewer than $K$ samples, sampling with replacement is used to maintain the batch structure.

Optimization is performed using AdamW with an initial learning rate $\eta = 10^{-3}$, weight decay, and maximum gradient norm clipping at 1.0. The learning rate follows a 10-epoch linear warmup followed by a cosine scheduler over 300 epochs (Luo et al., 2019). Automatic Mixed Precision (AMP) with floating-point 16 is enabled to optimize hardware utilization and training throughput.

### 4.3 HYBRID LOSS FUNCTION

Most ReID works combine an ID loss with a triplet loss to train the model (Luo et al., 2019). We follow this practice but optimize a broader hybrid objective that supervises both embedding learning and identity classification:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{ce}}\mathcal{L}_{\text{ce}} + \lambda_{\text{tri}}\mathcal{L}_{\text{tri}} + \lambda_{\text{sup}}\mathcal{L}_{\text{sup}} + \lambda_{\text{arc}}\mathcal{L}_{\text{arc}} \tag{5}$$

where $\mathcal{L}_{\text{ce}}$ is the cross-entropy loss, $\mathcal{L}_{\text{tri}}$ the triplet loss, $\mathcal{L}_{\text{sup}}$ the supervised contrastive loss, $\mathcal{L}_{\text{arc}}$ the ArcFace loss. We set all weights to one ($\lambda_{\text{ce}} = \lambda_{\text{tri}} = \lambda_{\text{sup}} = \lambda_{\text{arc}} = 1.0$) and leave their tuning to future work. The losses are detailed below.

**Label-Smoothed Cross-Entropy ($\mathcal{L}_{\text{ce}}$):** Identity classification uses CE with label smoothing ($\varepsilon = 0.1$) to avoid overconfident logits (Luo et al., 2019). The classification loss teaches to separate identities better whereas the metric learning losses focus on embedding separation.

$$\mathcal{L}_{\text{ce}} = -\sum_{i=1}^{N} q_i \log(p_i), \quad q_i = \left(1 - \tfrac{N-1}{N}\varepsilon\right)\mathbb{I}(y = i) + \tfrac{\varepsilon}{N}\mathbb{I}(y \neq i) \tag{6}$$

**Batch-Hard Triplet ($\mathcal{L}_{\text{tri}}$):** For each anchor $f_i$, the farthest positive and closest negative in the batch are selected to tighten the intra-class and enlarge inter-class cosine distances (Zeng et al., 2020):

$$\mathcal{L}_{\text{tri}} = \frac{1}{B}\sum_{i=1}^{B}\Big[\max_{j:y_j=y_i} d(f_i, f_j) - \min_{k:y_k\neq y_i} d(f_i, f_k) + m\Big]_+, \tag{7}$$

with margin $m = 0.3$.

**Supervised Contrastive ($\mathcal{L}_{\text{sup}}$):** Supervised contrastive loss draws each anchor toward every same-identity embedding in a batch, including those from other cameras, while simultaneously pushing it away from all other identities. This produces tighter and more compact identity clusters.

$$\mathcal{L}_{\text{sup}} = \sum_{i=1}^{B} \frac{-1}{|P(i)|} \sum_{p\in P(i)} \log \frac{\exp(f_i^\top f_p/\tau)}{\sum_{a\neq i}\exp(f_i^\top f_a/\tau)}, \tag{8}$$

where $P(i) = \{p \neq i : y_p = y_i\}$ and $\tau = 0.07$.

**ArcFace ($\mathcal{L}_{\text{arc}}$):** ArcFace tightens identity boundaries by making the correct class harder to reach. It adds an angular margin directly to the angle between an embedding and its true class prototype, lowering that class's logit unless the embedding moves even closer to the prototype:

$$\mathcal{L}\text{arc} = -\frac{1}{B}\sum i = 1^B \log \frac{\exp\big(s\cos(\theta_{i,y_i} + m_{\text{arc}})\big)}{\exp\big(s\cos(\theta_{i,y_i} + m_{\text{arc}})\big) + \sum_{k\neq y_i}\exp(s\cos\theta_{i,k})}, \tag{9}$$

with $\cos\theta_{i,k} = \frac{W_k^\top}{|W_k|_2} f_i$. This creates tighter same-identity clusters and better angular separation.

### 4.4 TRACKLET-LEVEL FEATURE AGGREGATION & EVALUATION STRATEGY

During inference, a tracklet $T = \{f_k\}_{k=1}^{K}$ containing $K$ frame-level embeddings $f_k \in \mathbb{R}^D$ is aggregated into a single tracklet-level descriptor $F_T \in \mathbb{R}^D$ for cross-camera identity matching. Embeddings are extracted for all query and gallery frames and tracklet descriptor embeddings are then generated using the selected aggregation strategy listed below.

**DBSCAN Density Clustering:** Density-based spatial clustering of all frame embeddings within a tracklet is done using cosine distance. Frames that are located in low-density regions are regarded as noise and are discarded. The primary dense cluster is identified, and the frame embeddings belonging to that cluster are averaged together to produce the tracklet descriptor.

**Generalized Mean (GeM) Pooling:** Generalizes mean and max pooling by applying a power-norm transformation ($p = 3.0$) to all frame embeddings, thereby emphasizing strong, repeated features across frames while suppressing weak background or occluded features. The sign vector preserves feature direction.

$$F_T = \operatorname{sgn}\left(\frac{1}{K}\sum_{k=1}^{K} f_k\right) \odot \left(\frac{1}{K}\sum_{k=1}^{K} |f_k|^p\right)^{1/p}. \tag{10}$$

Same object across different cameras is considered as positive and same object under same camera is considered as junk (Zheng et al., 2015). Finally, pairwise cross-camera matching between query and gallery tracklet descriptors is done by using Cosine Distance. Matching distance matrices are further refined using $k$-reciprocal re-ranking ($k_1 = 20, k_2 = 6, \lambda = 0.3$) to also consider local neighbourhood structure, rather than only comparing distance between two embeddings. Performance is evaluated using Rank-1 accuracy and mean Average Precision (mAP) ( B.3 ), wherein mAP serves as the primary metric.

# 5 Experiments & Results

In this section, we evaluate our models on Market-1501 and AI City / MTMC Tracking 2025. Table 2 compares our fine-tuned DINOv2 model against state-of-the-art person ReID methods on Market-1501. Unlike TransReID-SSL (Luo et al., 2021) and SOLIDER (Chen et al., 2023), which are self-supervised pre-trained on large-scale unlabeled person images (LUPerson), our backbone is pre-trained only on general-domain images (LVD-142M). Despite this domain gap, our fully trained DINOv2 achieves 96.6% Rank-1 and 93.1% mAP, without re-ranking. Adding $k$-reciprocal re-ranking raises mAP to **94.2%**, with a slight Rank-1 drop to 96.5% as k-reciprocal re-ranking promotes hard negatives over the true top match. Overall, our proposed method is competitive with self-supervised baselines on a standard real-world benchmark.

Table 2: Despite general-domain pretraining, our model remains competitive on Market-1501 with person-pretrained state-of-the-art methods such as SOLIDER and TransReID-SSL.

| Method | Rank-1 | mAP |
|---|---|---|
| TransReID-SSL | 96.7% | 93.2% |
| SOLIDER | **96.9%** | 93.9% |
| SOLIDER + Re-ranking | 96.7% | **95.6%** |
| DINOv2 (ours) | 96.6% | 93.1% |
| DINOv2 + Re-ranking (ours) | 96.5% | 94.2% |

## 5.1 Ablation Study of Proposed Pipeline

To analyze the impact of each component in the pipeline, step-by-step ablation studies were conducted on both Market-1501 and MTMC Tracking 2025 datasets. Note that we add components sequentially rather than independently in the experimental setup. We also do not run leave-one-out or re-weighted loss ablations because of limited GPU time.

**Market-1501 Ablation Study:** Table 3 shows the progressive addition of components to DINOv2 on Market-1501 to benchmark the pipeline on a standard dataset before evaluation on the AI City dataset. Pretrained DINOv2 without fine-tuning yields poor performance (0.8% mAP, 2.0% Rank-1), because of the domain gap between self-supervised representation and ReID retrieval. Introducing Batch-Hard Triplet Loss and full backbone fine-tuning lifts mAP to 88.9% and Rank-1 to 95.1%. Incorporating BNNeck, Random Erasing and hybrid losses steadily boosts performance to 92.3% mAP and 96.2% Rank-1. Combining CLS and Patch Average tokens, upgrading resolution to $384 \times 192$, and applying $k$-reciprocal re-ranking achieves near state-of-the-art performance (Table 2) of **94.2% mAP** and **96.5% Rank-1** with 100 epochs.

**MTMC Tracking 2025 Ablation Study:** Table 4 shows the method-by-method progression on our primary dataset (MTMC Tracking 2025) across both Image-Level and Tracklet-Level evalua-

Table 3: Ablation study of DINOv2 pipeline on Market-1501.

| **Configuration** | **Rank-1 (%)** | **mAP (%)** |
|---|---|---|
| `OSNet-AIN` (Pretrained on Market/MSMT/CUHK) | 93.4 | 82.2 |
| `OSNet` (Pretrained on Market-1501) | 94.3 | 83.6 |
| `DINOv2` (Pretrained on LVD-142M) | 2.0 | 0.8 |
| + BatchHard Triplet Loss (LoRA $r = 8$) | 93.3 | 86.5 |
| + BatchHard Triplet Loss (LoRA $r = 16$) | 94.2 | 87.6 |
| + BatchHard Triplet Loss (LoRA $r = 32$) | 93.9 | 86.8 |
| + BatchHard Triplet Loss (Full Fine-Tuning) | 95.1 | 88.9 |
| + BNNeck | 94.9 | 89.2 |
| + Random Erasing Augmentation | 95.0 | 90.3 |
| + Label-Smoothed Cross-Entropy ($\mathcal{L}_{\text{ce}}$) | 95.7 | 90.4 |
| + ArcFace Loss ($\mathcal{L}_{\text{arc}}$) | 95.9 | 91.7 |
| + Supervised Contrastive Loss ($\mathcal{L}_{\text{sup}}$) | 96.2 | 92.3 |
| + Hybrid Token Representation ($x_{\text{cls}}$ + Patch Avg) | 96.5 | 92.8 |
| + High Resolution ($384 \times 192$) | **96.6** | 93.1 |
| + $k$-Reciprocal Re-ranking | 96.5 | **94.2** |

Table 4: Ablation study of DINOv2 pipeline on MTMC Tracking 2025 Dataset.

| | **Image Level** | | **Tracklet Level** | |
|---|---|---|---|---|
| **Configuration** | **Rank-1 (%)** | **mAP (%)** | **Rank-1 (%)** | **mAP (%)** |
| `DINOv2` | 32.0 | 6.3 | 35.9 | 10.2 |
| + BatchHard Triplet (LoRA $r = 8$) | 88.1 | 70.9 | 95.4 | 81.2 |
| + BatchHard Triplet (LoRA $r = 16$) | 88.7 | 70.8 | 95.4 | 80.7 |
| + BatchHard Triplet (LoRA $r = 32$) | 79.1 | 46.2 | 82.0 | 56.2 |
| + BatchHard Triplet (Full Fine-Tuning) | 90.4 | 75.0 | 93.9 | 82.2 |
| + BNNeck | 89.7 | 75.7 | 95.4 | 80.7 |
| + Random Erasing | 89.7 | 76.0 | 93.9 | 81.1 |
| + Label-Smoothed CE | 92.2 | 79.8 | 93.9 | 85.8 |
| + ArcFace Loss | 92.4 | 81.3 | 97.0 | 87.5 |
| + SupCon Loss | 92.2 | 81.5 | 97.0 | 87.7 |
| + CLS + Patch Token Average | 91.7 | 81.6 | 97.0 | 87.5 |
| + High Resolution ($384 \times 192$) | 93.1 | 82.5 | 97.0 | 88.2 |
| + Extended Training (300 Epochs) | **93.7** | **84.5** | **97.0** | **90.0** |

tions, where the main focus is on the tracklet-level performance. The baseline GLANCE framework (Hashempoor, 2025) uses the pre-trained OSNet-AIN as the feature extractor backbone.

Pre-trained DINOv2 yields a very low baseline at image-level and tracklet-level. Introducing Batch-Hard Triplet Loss under LoRA ($r = 8$) improves tracklet mAP to 81.2%, while full fine-tuning further improves performance. Incorporating BNNeck and hybrid loss components progressively improves mAP to 87.7%. Finally, upgrading to high-resolution inputs ($384 \times 192$) and extending training to 300 epochs under full settings yields an Image-Level mAP / Rank-1 of **84.5% / 93.7%** and a Tracklet-Level mAP / Rank-1 of **90.0% / 97.0%**. As illustrated in Figure 1, our fine-tuned DINOv2 model successfully resolves identity confusions, even subtle variations in sleeve color and occlusions, significantly outperforming the baseline OSNet-AIN.

**Limitation:** DINOv2 ViT-B/14 processes 212 frames per second (FPS) on an A100 GPU, compared to 782 FPS for OSNet, a 3.7× slowdown. The +9.3 mAP gains come at a higher computational cost. We also observe two performance degradations. First, increasing the LoRA rank does not always improve performance. On MTMC Tracking 2025, $r = 8, 16$ reach about 71% image-level mAP, while $r = 32$ drops to 46.2%. The likely reason is that the learning rate and $\alpha$ were not re-tuned. A systematic study of LoRA hyperparameters is left for future work. Second, $k$-reciprocal re-ranking


Figure 1: Qualitative Re-ID performance for target ID 959 (orange hardhat, purple vest, light long sleeves). Green borders denote query and red indicates a false positive. Baseline OSNet-AIN fails at Rank 4 by confusing the query with a person wearing dark sleeves. Fine-tuned DINOv2 correctly retrieves all diverse top-5 matches despite partial occlusions, shifts, and viewpoint variations.

improves mAP on Market-1501 from 93.1% to 94.2% and on OSNet-AIN from 81.4% to 82.3% but lowers Rank-1, since a hard negative with a similar neighborhood can overtake an already-correct top match. The drop is small for DINOv2 on Market-1501 (96.6% to 96.5%) but larger for OSNet-AIN (96.2% to 93.1%), so we report mAP as the primary metric once re-ranking is applied.

### 5.2 Comparison of Tracklet Aggregation Methods

Table 5, compares the tracklet aggregation and post-processing strategies on MTMC Tracking 2025 using the fully trained DINOv2 backbone. DBSCAN density clustering achieves 90.0% mAP and 97.0% Rank-1. GeM Pooling ($p = 3.0$) effectively emphasizes discriminative features without completely ignoring weak features, thereby pushing mAP to 91.2% (with 95.4% Rank-1). Adding $k$-reciprocal re-ranking to GeM pooling yields the top performance of **91.6% mAP** and **97.0% Rank-1**. Several other methods like HDBSCAN and mean pooling (Section B.4) are also explored and any of these aggregation methods can be configured during evaluation.

Table 5: Comparison of tracklet aggregation methods on full-trained DINOv2.

| Aggregation Method | Tracklet Rank-1 (%) | Tracklet mAP (%) |
|---|---|---|
| DBSCAN Clustering ($\varepsilon = 0.2$) | 97.0 | 90.0 |
| HDBSCAN Clustering | 96.2 | 88.6 |
| Mean Pooling | 97.0 | 89.6 |
| Medoid Selection | 96.2 | 89.0 |
| Similarity-Weighted Pooling | 97.0 | 90.4 |
| Generalized Mean (GeM) Pooling | 95.4 | 91.2 |
| GeM Pooling + $k$-Reciprocal Re-ranking | **97.0** | **91.6** |

### 5.3 Analysis of Training Components

Table 6 presents an architectural comparison between OSNet, OSNet-AIN and the proposed DINOv2 pipeline under full experimental settings at the tracklet-level.

**DINOv2 as the Strongest Backbone:** Across every tracklet aggregation strategy, full-trained DINOv2 outperforms both CNN baselines. Under full fine-tuning with $k$-reciprocal re-ranking, DINOv2 reaches **91.6% mAP** (97.0% Rank-1), **+9.3 mAP** over the same OSNet and OSNet-AIN backbones fully trained with our recipe (both 82.3% mAP). Relative to the pre-trained OSNet-AIN + DBSCAN reference baseline that is not fine-tuned (54.3% mAP) the gap is +37.3 mAP.

Table 6: Architectural comparison of OSNet baselines and DINOv2 on AI City 2025. The red box denotes the baseline model used by Hashempoor (2025), while boldface indicates our results.

| **Model & Post-Processing** | **Rank-1 (%)** | **mAP (%)** |
|---|---|---|
| `OSNet` (pretrained on Market-1501) | 69.8% | 31.9% |
| → pretrained + DBSCAN | 87.0% | 55.4% |
| → Full trained + DBSCAN | 93.1% | 79.7% |
| → Full trained + GeM Pooling | 95.4% | 81.2% |
| → Full trained + GeM Pooling + Re-ranking | 95.4% | 82.3% |
| `OSNet-AIN` (pretrained on M, MS, C) | 71.3% | 33.3% |
| → pretrained + DBSCAN | 93.9% | 54.3% |
| → Full trained + DBSCAN | 95.4% | 79.0% |
| → Full trained + GeM Pooling | 96.2% | 81.4% |
| → Full trained + GeM Pooling + Re-ranking | 93.1% | 82.3% |
| `DINOv2` (pretrained on LVD-142M) | 32.0% | 6.3% |
| → pretrained + DBSCAN | 41.2% | 10.9% |
| → Full trained + DBSCAN | 97.0% | 90.0% |
| → Full trained + GeM Pooling | 95.4% | 91.2% |
| → Full trained + GeM Pooling + Re-ranking | **97.0%** | **91.6%** |

**Pre-trained Model Adaptation Gap:** Pre-trained DINOv2 displays low performance (6.3% mAP, 32.0% Rank-1) compared to pre-trained CNNs (`OSNet-AIN` at 33.3% mAP and `OSNet` at 31.9% mAP), highlighting that self-supervised ViT representations benefit substantially from task-specific supervised metric learning for fine-grained identity retrieval.

**Performance Improvement from GeM Pooling and Re-ranking:** Transitioning from DBSCAN clustering to GeM Pooling improves primary mAP across all three backbone families (`OSNet`: 79.7% to 81.2%, `OSNet-AIN`: 79.0% to 81.4%, DINOv2: 90.0% to 91.2%), and adding $k$-reciprocal re-ranking further pushes mAP to reach peak values across all backbones (`OSNet`: 82.3%, `OSNet-AIN`: 82.3%, DINOv2: 91.6%).

## 6 Conclusion

We presented a feature representation framework for multi-camera object re-identification based on the self-supervised Vision Transformer DINOv2. Combining the global class token with mean-pooled patch tokens captures both semantic context and local appearance, while a BNNeck, hybrid metric-learning loss, and Random Erasing produce compact embeddings that remain robust to appearance changes and partial occlusions. For tracklet-level matching, GeM pooling retains all frame-level features without the limitations of density-based clustering, and $k$-reciprocal re-ranking further refines retrieval.

On Market-1501, our pipeline reaches 94.2% mAP and 96.5% Rank-1, competitive to existing state-of-the-art pre-trained on LUPerson. On AI City / MTMC Tracking 2025 it reaches **91.6% mAP** and 97.0% Rank-1, **+9.3 mAP** over the same OSNet-AIN backbone fully trained with our recipe (82.3% mAP). Relative to the pre-trained OSNet-AIN + DBSCAN reference (54.3% mAP) the gap is +37.3 mAP, however, in this case the model is not fine-tuned. The largest gain therefore comes from the training recipe and tracklet aggregation, with the DINOv2 backbone further boosting performance.

This paper has two main limitations. First, DINOv2 ViT-B/14 runs at 212 FPS on an A100 GPU versus 782 FPS for OSNet, a 3.7× slowdown, so the +9.3 mAP gain comes at a latency cost. Second, AI City / MTMC Tracking 2025 provides no labeled ReID split, so we defined a Market-style protocol with 131 cross-camera identities and the longest tracklet as the query. The 91.6% mAP is therefore specific to this protocol and should not be read as an official AI City challenge score. Future work includes distilling the backbone for real-time inference, retuning the hybrid-loss weights, and self-supervised pre-training on the target domain, similar to SOLIDER and TransReID-SSL.

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# A LLM Usage

We used LLMs to assist with language editing, including grammar correction, sentence rephrasing, and improving clarity and readability throughout the report. LLMs were not used to design experiments or interpret results. All experimental design, training configurations, evaluation protocols, and analysis of results were carried out independently by the authors. The text was verified by the authors for correctness and accuracy prior to inclusion in this report.

# B Appendix

## B.1 Embedding Analysis

To evaluate the success of our framework against the primary objectives of fine-tuning a better feature extractor and improving feature selection beyond standard DBSCAN density clustering, we conduct an in-depth analysis of embedding quality, noise rates, and tracklet-level retrieval metrics. The following five metrics are used to examine embedding quality and clustering behaviour across backbone configurations in Table 7, while tracklet-level mAP and Rank-1 are used to assess the resulting retrieval performance in Table 8.

- **Same-ID Cosine Similarity**: Measures the similarity between the embeddings of the same identity across different cameras (higher value is better).
- **Diff-ID Cosine Similarity**: Measures similarity between embeddings of different identities (lower value is better).
- **Separation Gap**: The difference between Same-ID and Diff-ID similarity. A higher gap value indicates better separation between matching and non-matching identities.
- **Pair ROC-AUC**: Measures how accurately positive pairs are ranked above negative pairs, where higher values (closer to 1.0) indicates stronger discriminative ability.
- **Average Noise Rate**: The percentage of tracklets that are flagged as noise and are discarded by DBSCAN / HDBSCAN.

Table 7: Embedding Quality and Noise Analysis across Backbone Architectures and Settings.

| Model & Post-Processing | Same-ID Cosine Sim | Diff-ID Cosine Sim | Separation Gap | Pair ROC-AUC | Average Noise Rate |
|---|---|---|---|---|---|
| `OSNet` (pretrained on Market-1501) | 0.71 | 0.56 | 0.15 | 0.83 | – |
| → Pre-trained + HDBSCAN | 0.72 | 0.56 | 0.16 | 0.90 | 79.4% |
| → Pre-trained + DBSCAN | 0.73 | 0.56 | 0.17 | 0.90 | 32.7% |
| → Full Training + HDBSCAN | 0.65 | 0.04 | 0.61 | 0.99 | 84.2% |
| → Full Training + DBSCAN | 0.67 | 0.04 | 0.63 | 0.99 | 34.8% |
| → Full Training + GeM Pooling | 0.84 | 0.05 | 0.79 | 0.99 | – |
| `OSNet-AIN` (pretrained on M, MS, C) | 0.68 | 0.51 | 0.16 | 0.85 | – |
| → Pre-trained + HDBSCAN | 0.68 | 0.51 | 0.17 | 0.90 | 79.6% |
| → Pre-trained + DBSCAN | 0.70 | 0.51 | 0.19 | 0.90 | 40.9% |
| → Full Training + HDBSCAN | 0.62 | 0.05 | 0.56 | 0.99 | 84.2% |
| → Full Training + DBSCAN | 0.65 | 0.05 | 0.60 | 0.99 | 38.8% |
| → Full Training + GeM Pooling | 0.84 | 0.06 | 0.77 | 0.99 | – |
| `DINOv2` (pretrained on LVD-142M) | 0.42 | 0.32 | 0.10 | 0.69 | – |
| → Pre-trained + HDBSCAN | 0.41 | 0.31 | 0.10 | 0.71 | 83.0% |
| → Pre-trained + DBSCAN | 0.41 | 0.31 | 0.11 | 0.70 | 73.3% |
| → Full Training + HDBSCAN | 0.75 | 0.07 | 0.68 | 0.99 | 86.2% |
| → Full Training + DBSCAN | 0.76 | 0.07 | 0.70 | 0.99 | 27.9% |
| → Full Training + GeM Pooling | 0.91 | 0.08 | 0.83 | 1.0 | – |

**DBSCAN Analysis:** Following GLANCE (Hashempoor, 2025), we evaluate DBSCAN (distance threshold $\varepsilon = 0.2$, minimum samples $= 4$) over tracklet embeddings. As reported in Table 7 and Table 8, fine-tuned OSNet under DBSCAN reaches Same-ID $0.67$, Diff-ID $0.04$, and a separation gap of $0.63$, discarding $34.8\%$ of frames ($79.7\%$ mAP). Fine-tuned OSNet-AIN is similar to OSNet

Table 8: Extensive Ablation study across all backbones on tracklet-level results.

| Model and Post-Processing | Rank-1 | mAP |
|---|---|---|
| `OSNet` (pretrained on Market-1501) | 69.8% | 31.9% |
| → pretrained + HDBSCAN | 88.6% | 49.6% |
| → pretrained + DBSCAN | 87.0% | 55.4% |
| → Full trained + HDBSCAN | 93.9% | 78.0% |
| → Full trained + DBSCAN | 93.1% | 79.7% |
| → Full trained + GeM Pooling | 95.4% | 81.2% |
| → Full trained + GeM Pooling + Re-ranking | 95.4% | 82.3% |
| `OSNet-AIN` (pretrained on M, MS, C) | 71.3% | 33.3% |
| → pretrained + HDBSCAN | 91.6% | 49.7% |
| → pretrained + DBSCAN | 93.9% | 54.3% |
| → Full trained + HDBSCAN | 95.4% | 77.6% |
| → Full trained + DBSCAN | 95.4% | 79.0% |
| → Full trained + GeM Pooling | 96.2% | 81.4% |
| → Full trained + GeM Pooling + Re-ranking | 93.1% | 82.3% |
| `DINOv2` (pretrained on LVD-142M) | 32.0% | 6.3% |
| → pretrained + HDBSCAN | 47.3% | 12.1% |
| → pretrained + DBSCAN | 41.2% | 10.9% |
| → Full trained + HDBSCAN | 96.2% | 88.6% |
| → Full trained + DBSCAN | 97.0% | 90.0% |
| → Full trained + GeM Pooling | 95.4% | 91.2% |
| → Full trained + GeM Pooling + Re-ranking | **97.0%** | **91.6%** |

(Same-ID $0.65$, Diff-ID $0.05$, gap $0.60$, noise $38.8\%$, $79.0\%$ mAP). Fine-tuned DINOv2 under DBSCAN improves Same-ID to $0.76$ and the gap to $0.70$, while rejecting only $27.9\%$ of frames, and reaches $90.0\%$ mAP ($97.0\%$ Rank-1). Pair ROC-AUC is $0.99$ for every fully trained DBSCAN row, so we do not use it to rank methods.

**HDBSCAN Analysis:** To improve upon DBSCAN, we also explored HDBSCAN to discover clusters without a fixed distance threshold $\varepsilon$. After complete training, it detects $84.2\%$ of frames as noise in both OSNet variants and $86.2\%$ in DINOv2. That extra rejection slightly lowers Same-ID relative to DBSCAN (OSNet $0.65$ vs. $0.67$, OSNet-AIN $0.62$ vs. $0.65$, DINOv2 $0.75$ vs. $0.76$) and reduces mAP on every backbone ($79.7\%$ to $78.0\%$ for OSNet, $79.0\%$ to $77.6\%$ for OSNet-AIN, $90.0\%$ to $88.6\%$ for DINOv2). DBSCAN is the better clustering choice here because it keeps more inliers.

**GeM Pooling Comparison:** Replacing clustering with GeM pooling ($p = 3.0$) keeps every frame and widens the gap further. OSNet reaches Same-ID $0.84$, Diff-ID $0.05$, and gap $0.79$ ($81.2\%$ mAP); OSNet-AIN reaches Same-ID $0.84$, Diff-ID $0.06$, and gap $0.77$ ($81.4\%$ mAP). DINOv2 + GeM is strongest: Same-ID $\mathbf{0.91}$, Diff-ID $0.08$, gap $\mathbf{0.83}$, i.e. **91.2% mAP** (**91.6% mAP** with $k$-reciprocal re-ranking).

### B.2 2D Visualization – UMAP

As shown in Figure 2, the baseline OSNet features exhibit a dense embedding region in which gallery samples (blue points) and query instances (red triangles) substantially overlap. This overlap indicates weak inter-class separation and feature ambiguity due to the lack of Re-ID specific training.

In contrast, Figures 3 and 4 show that the fine-tuned models produce more structured and better-separated embedding spaces. In particular, the DINOv2 embeddings has query instances close to compact and relatively tight gallery clusters, suggesting improved identity discrimination and clearer cluster boundaries.

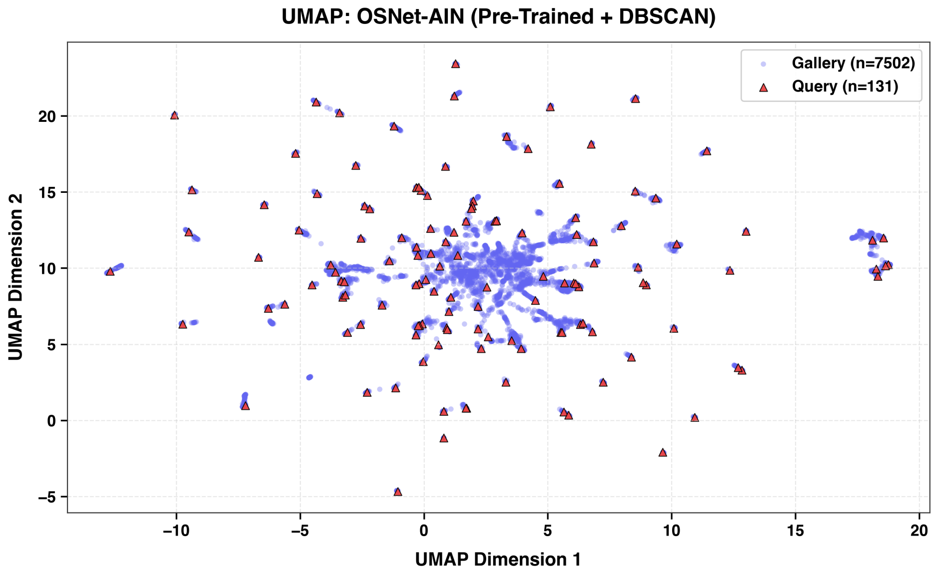


Figure 2: UMAP of pretrained OSNet-AIN (`osnet_ain_x1_0`) + DBSCAN at tracklet-level. Gallery embeddings are scattered with respect to the queries as the model is not fine-tuned.

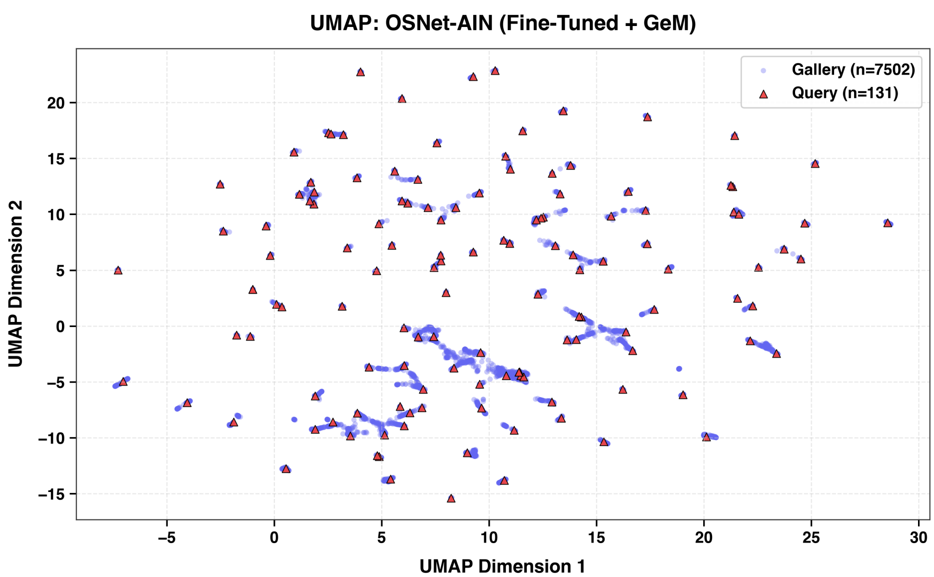


Figure 3: UMAP of fully trained OSNet-AIN (`osnet_ain_x1_0`) + GeM at tracklet-level. Gallery embeddings are forms clusters near the queries when the model is trained with AI City dataset.

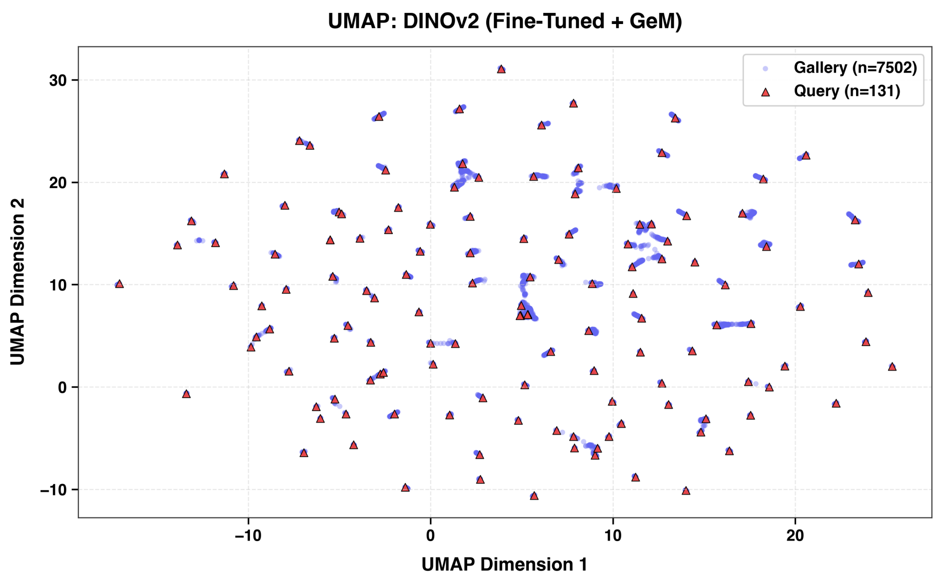


Figure 4: UMAP of fully trained DINOv2 (`dinov2_vitb14`) + GeM at tracklet-level. Gallery embeddings are tightly clustered around the queries due to stronger embeddings from DINOv2.

### B.3 Evaluation Metrics

Performance is evaluated using Rank-1 accuracy and Mean Average Precision (mAP).

**Rank-1 Accuracy:** Measures the percentage of queries where the top-1 retrieved match is correct:

$$\text{Rank-}k = \frac{1}{N_q} \sum_{i=1}^{N_q} \mathbb{I}\left( \min_{j \in \mathcal{M}(q_i)} \text{rank}(g_j) \leq k \right), \tag{11}$$

where $k$ = 1 and $\mathcal{M}(q_i) = \{j : y_{g_j} = y_{q_i} \wedge c_{g_j} \neq c_{q_i}\}$ contains valid positive gallery indices for query $q_i$.

**Mean Average Precision (mAP):** Measures retrieval precision and ranking quality across all correct matches. For query $q_i$, Average Precision (AP$_i$) and mAP across $N_q$ queries are defined as:

$$\text{AP}_i = \frac{\sum_{k=1}^{N_g} P(k) \cdot \delta(k)}{N_{\text{pos},i}}, \quad \text{mAP} = \frac{1}{N_q} \sum_{i=1}^{N_q} \text{AP}_i, \tag{12}$$

where $P(k)$ is the precision at rank $k$, $\delta(k)$ indicates non-junk positive matches, and $N_{\text{pos},i} = |\mathcal{M}(q_i)|$.

### B.4 Other Tracklet Aggregation Methods

**Mean Pooling** Each feature is independently averaged to generate the aggregated feature vector and each frame vector has same uniform importance.

$$F_T = \frac{1}{K} \sum_{k=1}^{K} f_k \tag{13}$$

**Similarity-Weighted Pooling:** Each frame $f_i$ calculates the pairwise cosine similarity between each frame embedding $f_i$ and all other frames $f_j$ in the tracklet.

$$c_i = \frac{1}{K} \sum_{j=1}^{K} \cos(f_i, f_j) \tag{14}$$

These scores are passed through a temperature-scaled softmax ($T_{\text{weight}} = 10.0$) to produce normalized weights $w_i$ and the tracklet descriptor $F_T$ is formed by aggregating all frame embeddings according to their assigned weights:

$$w_i = \frac{\exp((c_i - \max_j c_j) \cdot T_{\text{weight}})}{\sum_{k=1}^{K} \exp((c_k - \max_j c_j) \cdot T_{\text{weight}})}, \quad F_T = \sum_{i=1}^{K} w_i f_i. \tag{15}$$

**Medoid Selection:** The average pairwise cosine similarity $c_i$ is calculated and the frame embedding that has the highest average cosine similarity to all other frames in the tracklet is selected.

$$k^* = \arg \max_{i \in \{1,\ldots,K\}} (c_i), \quad F_T = f_{k^*}. \tag{16}$$

**HDBSCAN Hierarchical Clustering:** An evolution of DBSCAN that discovers clusters across varying densities (minimum cluster size $= 4$) without requiring a fixed distance threshold $\varepsilon$. It automatically discards outlier frames and applies mean-pooling to the primary cluster embeddings to create the final tracklet descriptor.